Literature Review & Research Gap Analysis

Topic: **LLM Multi-Agent Collaboration** | Generated: 2026-10-21 10:38:02 UTC

1. Search Methodology & Retrieval Overview

A systematic multi-source literature retrieval was conducted for the topic 'LLM Multi-Agent Collaboration'. Scholarly records were extracted from open APIs (OpenAlex and arXiv, with Semantic Scholar where available). Deduplication was executed across normalized DOIs, arXiv IDs, and sequence title similarities. Retrieved publications were scored by title/abstract keyword relevance combined with publication recency. Each paper was independently decomposed into 9 structural dimensions. Evidence grounding was audited to verify that all gap conclusions correlate with verifiable statements from the literature.

2. Analyzed Academic Papers

1. GUARDIAN: Safeguarding LLM Multi-Agent Collaborations with Temporal Graph Modeling

Problem: However, multi-agent collaboration faces critical safety challenges, such as hallucination amplification and error injection and propagation.

Methodology: Not reported in the available text.

Models/Algorithms: LLM

Datasets: GitHub

2. Poster: LLM Multi-Agent Collaboration for Network Deployment and Management

Problem: This paper presents the MAPLE framework, which harnesses large language models (LLMs) to facilitate multi-agent collaboration for fully automated deployment and management of large-scale networks.

Methodology: Not reported in the available text.

Models/Algorithms: Not reported in the available text.

Datasets: Not reported in the available text.

3. Audit-LLM: Multi-Agent Collaboration for Log-based Insider Threat Detection

Problem: Log-based insider threat detection (ITD) detects malicious user activities by auditing log entries.

Methodology: To enhance conclusion accuracy, we propose a pair-wise Evidence-based Multi-agent Debate (EMAD) mechanism, where two independent Executors iteratively refine their conclusions through reasoning exchange to reach a consensus.

Models/Algorithms: LLM, Chain-of-Thought

Datasets: ITD

3. Potential Research Gaps & Grounding Evidence

Gap 1: Handling Dynamic Runtime Errors and Environmental Feedback in LLM Multi-Agent Collaboration (Confidence: Low)

Category: real_world_deployment | **Type:** explicit_author_limitation

Description: Analyzed architectures frequently assume static or simulated environments. When encountering dynamic runtime exceptions, agents struggle with cascading errors and lack robust self-healing feedback mechanisms.

Significance: Autonomous systems cannot operate safely in production without fail-safe error recovery and deterministic verification.

Supporting Evidence:

- *GUARDIAN: Safeguarding LLM Multi-Agent Collaborations with Temporal Graph Modeling*: "However, multi-agent collaboration faces critical safety challenges, such as hallucination amplification and error injection and propagation."

Gap 2: Cascading Hallucination and Dialogue Stagnation in Collaborative LLM Multi-Agent Collaboration (Confidence: Low)

Category: algorithm | **Type:** explicit_author_limitation

Description: Inter-agent conversation chains are susceptible to compounding hallucinations, where an early error by one agent propagates unchecked through downstream collaborative steps.

Significance: Error accumulation undermines system trustworthiness and leads to catastrophic failures in complex, multi-stage workflows.

Supporting Evidence:

- *GUARDIAN: Safeguarding LLM Multi-Agent Collaborations with Temporal Graph Modeling*: "However, multi-agent collaboration faces critical safety challenges, such as hallucination amplification and error injection and propagation."

Gap 3: Potential Gap in Standardized & Multi-Domain Evaluation Benchmarks for LLM Multi-Agent Collaboration (Confidence: Medium)

Category: dataset | **Type:** inferred_cross_paper

Description: Cross-paper synthesis reveals that analyzed systems predominantly rely on restricted synthetic tasks or a narrow set of code/text benchmarks. Standardized real-world stress-testing benchmarks remain underexplored.

Significance: Without unified benchmark suites capturing multi-step dynamics, evaluating comparative progress and reproducibility across independent agent architectures remains problematic.

Supporting Evidence:

- *GUARDIAN: Safeguarding LLM Multi-Agent Collaborations with Temporal Graph Modeling*: "Datasets utilized: GitHub"
- *Poster: LLM Multi-Agent Collaboration for Network Deployment and Management*: "No standardized public benchmark reported."
- *Audit-LLM: Multi-Agent Collaboration for Log-based Insider Threat Detection*: "Datasets utilized: ITD"

4. Bibliographic References

1. Jialong Zhou, Lichao Wang, Xiao Yang (2025). *GUARDIAN: Safeguarding LLM Multi-Agent Collaborations with Temporal Graph Modeling*. neural information processing systems.
2. Zhengyi Cheng, Chongxi Ma, Mingxuan Tang et al. (2025). *Poster: LLM Multi-Agent Collaboration for Network Deployment and Management*. Proceedings/Proceedings - International Conference on Network Protocols.
3. Chengyu Song, Linru Ma, Zheng, Jianming et al. (2024). *Audit-LLM: Multi-Agent Collaboration for Log-based Insider Threat Detection*. arXiv (Cornell University).